# GHASSAN ARNOUK

**Bachelor of Engineering, Electrical | Co-operative Education**

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## EXPERIENCE

### Junior Programming Developer [Casual]

**Public Service Commission of Canada**

Jul 2022 – Nov 2022 | Gatineau, Quebec

- **Resolved software bugs**, using **Java** and **SQL**, for both **PSRS** and **PIMS applications** to meet internal customer design specifications

### Teaching Assistant

**Carleton University**

Jan 2022 – Apr 2022 | Ottawa, Ontario

- Helped students **understand and master calculus** topics by **breaking down concepts** during weekly tutorials

### Software Engineering Co-op

**MDS Aero Support Corporation**

May 2021 – Dec 2021 | Ottawa, Ontario

- **Designed and built** phase I of the **automated testing framework**, using **bash** & **python**, reducing manual testing effort by finding defects in the product early in the development cycle
- Worked in an **agile environment** using **JIRA** and **Confluence**
- **Mentored new** team **members** through daily training sessions

### Product Verification Engineering Co-op

**MDS Aero Support Corporation**

Sep 2020 – Apr 2021 | Ottawa, Ontario

- Performed **manual testing** of the company's Data Acquisition System (DAS) **to ensure software** & hardware **meet** internal & customer **design specifications**
- **Raised** and **verified** software bugs in **Bugzilla** before each release
- **Designed** and **executed** Acceptance Test Procedures (ATPs) per nxDAS release in **TestLink**

### Research Assistant

**Carleton University**

Jul 2020 – Dec 2020 | Ottawa, Ontario

- Evaluated the strengths and weaknesses of various rectifier circuits by **running simulations** in **LTSpice**
- Created a **PCB prototype** of the finalized circuit design in **KiCad**
- **Delivered a report** detailing the completed work and all generated simulation and CAD files in LaTeX

## SKILLS

- **PROGRAMMING LANGUAGES**
  **Proficient:** Bash | Python | C/C++ | LaTeX
  **Prior Experience:** Verilog | Java | SQL | Assembly
- **SOFTWARE DEVELOPMENT**
  Linux | Regex | Git | GitLab | GitHub | Ansible | SVN | Keil
  Vim | VS Code | Emacs | IntelliJ
- **HARDWARE**
  FPGA | Raspberry Pi | MKL25Z
  LTSpice | KiCad | Altium 18 | Soldering

## PROJECT ACTIVITIES

### Autonomous Vehicles

**Team Member, Capstone Project**

Sep 2022 – Present

- Performed self-directed learning of **ARM-based embedded system** concepts such as **assembly**, **interrupts**, **PWM** & **DAC**, **ADC** & **I2C**, **SPI**, **UART**
- Introduced a **motor control circuit**, operated **using C code**, to **control** motor **speed** via a **PWM** signal

### Smart Solar Panel System

**Team Leader, Third-Year Project**

Jan 2022 – Apr 2022

- **Led the team** in **designing** & **building** an improved **solar system** used to track the movement of light
- An **algorithm** was implemented, in **Python**, to **track light** & **switch mode** between automatic & manual.
- A **GUI** was developed, using **PyQt5 framework**, to **display data** and provide convenient **switching between manual and automatic** modes
- Filmed & **produced a video** of the finalized project

### SOMETHING GOES HERE

**Personal Project**

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### IEEE Carleton

**Treasurer & Workshops Presenter**

Sep 2020 – Present

- **Performed budget planning** for events and workshops in addition to organizing **SPAC 2022** finances
- **Running** 4-series LaTeX **workshops** for students **demonstrating** various **development** and automation **environments** using **vim**, **vs code**, and **emacs**.

### Carleton Planetary Robotics Team

**Power Team Member**

Sep 2018 – Apr 2020

- **Designed** and **built** an improved power distribution board (**PDB**) in **Altium Designer 18** for a mars rover

## HOBBIES & LANGUAGES

- **Basketball** (GSW fan), Cycling, Spikeball, and Yoga
- Fluent in both **Arabic** and **English**